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Priming Effect of Colour on Aiding the Attentional Reorientation in Sequential Presentations of Temporal Data Visualization: Evidence From Eye-Tracking

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In the present study, we focus on the priming effect of colour on mitigating the attentional reorientation cost, which is led by re-constructing the frame of reference for attention shift and visual search in sequential presentations of temporal data visualization. The study involves two experiments using complementary recordings of behavioural performance and eye-tracking events. Two aspects of colour primes are highlighted: the prime validity and the colour perceptual accessibility. A task paradigm integrating the feature search and keeping-track task was adopted in our experiments. In Experiment 1 (with a group of 16 participants), we confirmed the colour priming effect by comparing the priming condition to the neutral baseline. Furthermore, global colours that are with high perceptual accessibility generated more evident priming effects than local colours. However, more interferences in misguiding the attention to task-irrelevant regions were found when the global primes were invalid. In Experiment 2 (with another group of 15 participants), we verify the finding in Experiment 1 that global colours produced more pronounced priming effects in alleviating the attentional reorientation cost by comparing two groups of real-world visualizations with either global or local colours as the prime. Large saccades were initiated much earlier, and the search efficiency got improved when provided with global colours. We conjecture that the facilitatory effect from global colours may stem from its benefit on the pre-attentive processing of the search field. The research findings provide evidence for utilizing colours as the primes in mitigating the attentional reorientation cost and accelerating visual search in sequential presentations.

RESEARCH HIGHLIGHTS

- Complementary recordings of behavioural and eye-tracking measures confirm the priming effect of colour on guiding the attention more efficiently after the presentation replacement.
- High colour perceptual accessibility (i.e. global colours) intensifies its priming effect and improves the search efficiency over low accessible colours (i.e. local colours).
- More interference is found when the global colour primes were invalid, whereas the interference effect was not significant when providing the invalid local primes.

Keywords: visualization design and evaluation methods; laboratory experiments; eye-tracking technology; attentional reorientation; priming effect.

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1. INTRODUCTION

The concept of temporal data denotes the evolution of state or object characteristics over some time. Due to the entanglement between time and space, the temporal data visualization (*TemVis*) often delineates the variation of dependent variables in the conceptual and topological aspect along the time and space dimension (Blok, 2000). *TemVis* plays a key role in supporting visual analytics and maintaining situational awareness in the domain of meteorological data analytics (Zhang *et al.*, 2019), co-authorship network exploration (Jing *et al.*, 2019), ecological migration tracking (Konzack *et al.*, 2018), urban mobility pattern extraction (Senaratne *et al.*, 2018), medical recording inspection (Gotz and Stavropoulos, 2014; Wongsuphasawat and Gotz, 2012), etc.

Among previous research concerning temporal data visualizations, there are two mainstream research foci. Some focused on *visual mapping* (Patterson *et al.*, 2014), which is also termed as the *graph structural dimension* in the *TemVis* (Kerracher *et al.*, 2014). It refers to representing relations between elements in a readable manner, namely using the radial layout to visualize periodic time (Pu *et al.*, 2013) or the linear layout to visualize linear time (Reda *et al.*, 2011), etc. Meanwhile, some focused on *view presentation* (Patterson *et al.*, 2014) that denotes how graph structures are presented to the user. One pivotal issue in the view presentation is determining how the *temporal dimension* is mapped to space (static) or time (dynamic) and the dimensionality of the presentation space (2D or 3D) (Kerracher *et al.*, 2014). In existing *TemVis*, we categorize the temporal encoding method into two genres: one encoded by *multiple time slices*, the other being *embedded* within a single graph structure. The concept of *time slice* originates from the conceptual representation of space–time cube (Bach *et al.*, 2014). Multiple time slices can either be presented sequentially (Archambault *et al.*, 2011; Archambault and Purchase, 2013; Archambault and Purchase, 2016), simultaneously (Albo *et al.*, 2016) or in a hybrid manner (Bach *et al.*, 2015). The sequential presentation of time slices in *TemVis* often entails the total replacement of the current presentation, which becomes the background of the current research. In Fig. 1, we visualize the relationship between concepts in *TemVis*.

There is a growing tendency to present all time slices simultaneously onto a single display. However, it will inevitably lead to information overwhelming, display clutter and additional interaction load (e.g. zoom in, drag) that will impair the information foraging performance (Neider and Zelinsky, 2011; Ognjanovic *et al.*, 2019). By contrast, detailed information of each time slice can be shown much clearly, and display clutter can be optimized in sequential presentations. Furthermore, considering the time-to-time mapping of the data to visualization, sequential view presentations provide more cognitive affordance to users when exploring the evolution of temporal events (Hartson, 2003). The predominance of sequential presentation of *TemVis* promotes its application in the pattern mining of geospatial information (Acevedo and Masuoka, 1997; Ogao and Kraak, 2002), anomaly detection in the power grid (Gegner *et al.*, 2016) and air traffic control (Hurter *et al.*, 2008).

However, sequential presentations require frequent *view-to-view transition*. Specifically, upon reaching the new view (or presentation), the viewer needs to decide where they should land on their first attention spot before the inspection (Corbetta *et al.*, 2008). The process involves breaking the attentional inertia onto the preceding presentation and reorienting the attention in the new viewport. Therefore, it consumes limited cognitive resources and brings additional cost (we term it as *attentional reorientation cost* below). Given this, the focus of the current study is how to mitigate the attentional reorientation cost in the sequential presentation of *TemVis*. Here, we choose the *colour* as the mitigating factor. The facilitatory effect of colour has been confirmed on object identification (Treisman, 1986), memory retrieval (Hanna and Remington, 1996) and emotion arousal (Wilms and Oberfeld, 2018) in visualization designs. However, it remains unknown to what extent *colour* can act as a *prime* in aiding the initial attentional decision in sequential presentations. Throughout the study, we seek to answer the following research questions:

RQ1: How to represent the attentional reorientation cost in sequential presentations of *TemVis* by eye movement measures?

RQ2: Whether the colour can take its priming effect in mitigating the attentional reorientation cost?

RQ3: If ***RQ2*** is confirmed, how the perceptual accessibility of colours and prime validity interact to influence the priming effect?

The paper is structured as follows: prior related work is discussed in section 2. The experimental method is discussed in section 3, which starts with the general methodology of two experiments. In section 4 and 5, two experiments are introduced in sequence. General discussion of two experiments and conclusion are established in section 6 and 7, respectively.

2. RELATED WORK

2.1. The role of pre-attentive and attentive processing in information visualizations

Information visualizations are often crowded with sensory impressions that exceed what the brain can process simultaneously. Selecting the most relevant information for processing while filtering out less relevant or irrelevant information is critical for information foraging in visualizations. The selection mechanism is referred to as *attention* (Desimone and Duncan, 1995). A two-stage visual attentive processing model was proposed: the early *pre-attentive processing*, which is effortless and automatic, and the later *attentive processing* that involves focal attention to put features together within the current attentional spotlight (Neisser, 2014a; Treisman and Gelade, 1980). The pre-attentive process is conjectured to relate to the scene perception and the detection of significant events in

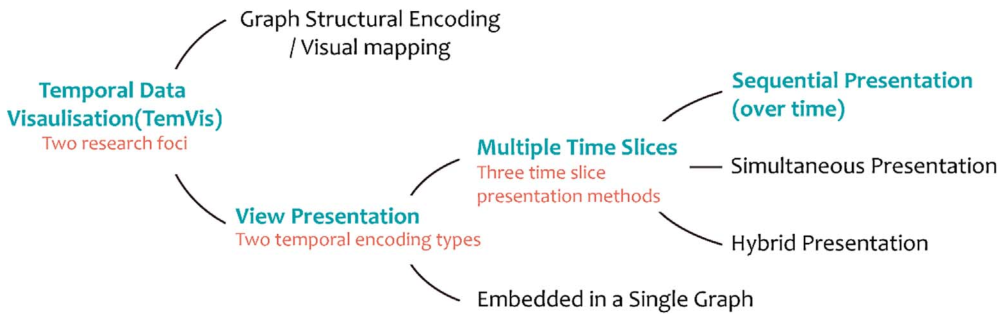


FIGURE 1. The hierarchical concepts of *TemVis* in the current study.

the peripheral vision, while the attentive process relates to fine discrimination (Trevarthen, 1968). The two stages take place sequentially (Bundesen, 1990; Theeuwes, 1993), wherein the pre-attentive mechanism generates a subjective primal sketch that is stored in the iconic memory (Neisser, 2014b). The primal sketch then permits the viewer to allocate more cognitive resources to the most relevant part through selective attention to gain a greater 'figural emphasis' (Broadbent, 1971; Kahneman, 1973). In the pre-attentive processing, attention can either be captured by the salience of stimulus (i.e. through the bottom-up processing), by the viewer's intention or will (i.e. top-down selection) or by the interaction between the two. This brings the distinction of *pure capture* and *contingent capture* (Folk and Remington, 2006). In the current study, the *priming effect* of colours relies on the *contingent capture* in pre-attentive processing. The colour that has been previously seen is expected to guide the attention to the same or similar colours in the subsequent scene. In information visualizations, previous work employed closure and depth (Perera *et al.*, 2007), hue and orientation (Healey *et al.*, 1996; Kunze *et al.*, 2018), sharpness (Kosara *et al.*, 2002) and brush stroke (Healey *et al.*, 2004) as the pre-attentive feature to attract user's attention. However, pre-attentive features can also distract the attention to salient but irrelevant areas (Li and Moacdieh, 2014). If such features are properly designed, they can support the attentional reorientation process that underlies the subsequent in-depth object processing. The attentive processing of objects lies at the heart of the *visual search* paradigm, in which the viewer needs to judge whether a designated target is present or not (Bravo and Nakayama, 1992; Stolte *et al.*, 2020).

2.2. Attentional reorientation in sequential presentations

The term 'reorientation' originates from the domain of spatial cognition and navigation. It denotes re-establishing one's allocentric position after disruption or disorientation (Cheng, 1986), which is the premise of performing the goal-directed behaviour. Here, we build a conceptual metaphor between skipping to a new viewport in the sequential presentations and physically entering a new space. From the perspective of the entire search process, Castelhano *et al.* (2008) divided the visual search into two independent epochs: target latency and target verification. The viewer is to locate the target during the target latency and decide whether the fixated object is the target during the target verification. Founded on this, Malcolm and Henderson (2009) extracted three functional epochs of visual search: search initiation, scanning and verification. On the other side, each spatial attention unit can be modelled as attention orientating and attention maintenance (Müller and Rabbitt, 1989). Posner and Presti further conceptualized the orienting process into three sub-processes: disengagement, shifting and engagement (1987). Supposing the synchronization between eye fixation and attention focus, the viewer needs to program the saccadic eye movement before orienting their attention focus and conducting the eye movement, which will prolong the response latency to the new attentional set. Delayed response latency was found in a dichotic selective listening task (Gopher, 1973). As well, discontinuity of a coherent auditory stream prolongs the response latency to relevant stimuli in the first few responses, which is described as a *cost* of interference (Lim *et al.*, 2019; Shinn-Cunningham, 2008). In vision-based research, fixation duration increased gradually over time while saccadic amplitude decreases simultaneously following the new scene. The two metrics reach a plateau in later viewing (Antes, 1974; Unema *et al.*, 2005). The shorter fixation pauses are conceived as a proxy for adaptation to the new scene, reflecting the process of re-setting up the attention according to the new spatial configuration (Karpov *et al.*, 1968). Also, delayed behavioural latencies were observed when navigating to a visual presentation area (Kortschot *et al.*, 2018). Therefore, akin to selective auditory attention and scene perception, we presume that the view-to-view transition can also lead to additional demands for reorienting the attention in the new presentation. In the present study, we divided the visual search into two behaviorally defined epochs: *reorientation* and *viewing*. Here, we define *reorientation* as the process of mentally re-setting up one's attention flow after the view transition and programming the subsequent saccadic eye movements. Visual

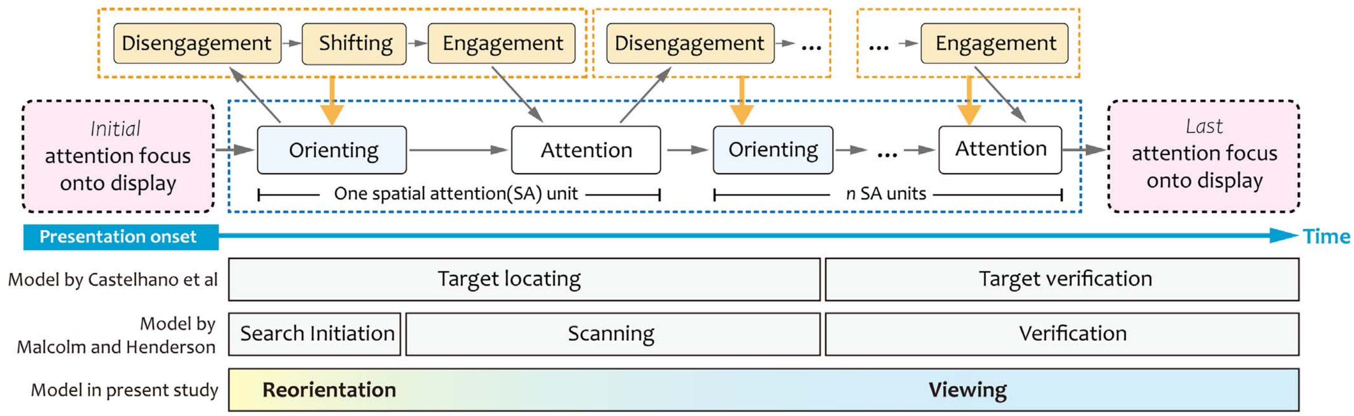


FIGURE 2. The attention flow from the onset of presentation. The figure shows the relation between the process of reorientation, the two mechanisms proposed by Müller and Rabbitt (1989) and the three stages of orienting by Posner and Presti (1987). Note that the chromatic gradient bar on the bottom represents the gradual transition from reorientation to regular viewing. We did not find supportive evidence for a clear boundary between the two epochs. There is no quantitative meaning of the colour proportion on the gradient bar.

search in this process should be less efficient than the following viewing epoch. In Fig. 2, we compared our visual search epoch division with the past studies. To our knowledge, few if any research discussed the reorientation process after presentation replacement in sequential *TemVis*. It was found that chromatic colours enhanced the discrimination of relevant information and contributed to the attentional guidance after video editorial cuts (Valuch and Ansorge, 2015). Furthermore, both visual similarity and matching properties enable a rapid attention deployment following a cut (Seywerth *et al.*, 2017; Valuch *et al.*, 2017).

2.3. Priming effect on attentional guidance

Priming effect refers to the facilitation effect on the processing of the current stimulus by the preceding similar or related stimulus (i.e. the prime) (Van den Bussche *et al.*, 2010). In visual search studies, the prime can act as a pre-cue to guide the attention during the orienting process (Jonides, 1983; Müller and Rabbitt, 1989). Both behavioural and eye movement evidence in the easy target detection task support the priming effect from perceptual primes (e.g. colour) (Andersen and Maier, 2019; Biggs *et al.*, 2015; Maljkovic and Nakayama, 1994; Nordfang *et al.*, 2013), symbolic primes (e.g. arrows) (Green and Woldorff, 2012; Ristic *et al.*, 2007; Tipples, 2002), locational primes (Chun and Jiang, 1998; Tipper *et al.*, 1990) and conceptual primes (e.g. semantically or syntactically related objects) (Castelhano and Heaven, 2011; Malcolm and Henderson, 2010). It was believed that perceptual priming takes effect in the pre-attentive processing stage and it relates to a perceptual identification process in the perceptual representation system, whereas conceptual priming engages semantic memory of referents (Tulving and Schacter, 1990). In general, faster detection speed, fewer errors and shorter first saccadic latency were observed when the prime and the target that is subsequently presented is congruent (we term as the *valid prime* in this study) than in the uninformative and incongruent priming condition (we term as the *invalid prime*) (Becker, 2008; Maljkovic and Nakayama, 1994; McPeek *et al.*, 1999; Schacter *et al.*, 1990; Soto and Humphreys, 2007). The attentional bias and guidance from primes were conceived to stem from the object or template held in the *working memory* (WM). The WM content exerts a tuning effect in early vision (Downing, 2000; Soto *et al.*, 2008). However, the priming effect was confirmed in easy target detection tasks in previous work, where the target has a high discernibility among distractors. Few, if any, examined this effect in complex visual search scenarios, for example, when the target is embedded in the display with a high visual clutter, or scrutiny of each search alternatives are involved in identifying the target. These inspire us to study the priming effect of colour by integrating the feature search with the keeping-track task paradigm.

2.4. Keeping-track task paradigm

Keeping-track task paradigm refers to the scenario where the viewer needs to monitor the updating information, encode the information into the WM and then make a judgment concerning the variation of the target over a short time interval (i.e. increase, decrease or remain unchanged) (Hess *et al.*, 1999). The task is typical in the intensive-care unit and common-control room. In these control centres, following a changing situation and keeping it in mind is pivotal for maintaining situational awareness. Yntema and his colleagues first developed the keeping-track paradigm. In their studies, participants were instructed to track the current state of many attributes of one object or of one attribute of many objects and then recall one after the interference. They found that the keeping track performance was related to the volume of possible state set and the frequency of changing (Yntema, 1963; Yntema and Mueser, 1962). Hess *et al.* extended Yntema's work and demonstrated that invariant display configuration could support the WM management of changing information (Hess *et al.*, 1999). By presenting the information serially, previous work emphasized how the WM management ability was affected by external factors. Therefore, they cut down on the task complexity from looking for the potential target. The current study integrated a conjunctive visual search task component where the viewer needed to locate and identify the target among distractors and encode it into memory. The combination of conjunctive visual search and keeping track task makes our task setting much closer to the real-world scenario and distinct from previous visual priming studies.

2.5. Reasons for choosing the colour as the prime

Previous work in experimental psychology corroborated the importance of colour in visual attention. For one thing, colours can summon attention automatically (Nothdurft, 1993; Turatto and Galfano, 2000). Visual search is facilitated for coloured targets and interfered with by coloured distractors (Biggs *et al.*, 2015; Müller *et al.*, 2009; Nordfang *et al.*, 2013). In feature search displays, coloured objects are more easily discerned among distractors than shapes (Lee *et al.*, 2018) and orientation (Sobel *et al.*, 2009). For another, the automatic attentional capture by colours can be manifested as attentional guidance. The attentional guidance performance varies across colours (Andersen and Maier, 2019) and the visual load in the display (Biggs *et al.*, 2015). In our prior work, we have confirmed the cueing effect of chromatic colours in improving the correspondence across viewports presented simultaneously in multiple-view visualizations (Peng *et al.*, 2021). As follow-up research, we intend to investigate how colour plays a role in the sequential presentation environment in this study. Moreover, in previous research, the designated object and the colour feature is either spatially integrated (i.e. colour fell on the object surface) or embedded (i.e. colour formed the background of the object) (Lloyd-Jones and Nakabayashi, 2009), thus leading to the categorization of *local* and *global* colours in the display. Global colours are visible in peripheral vision (Holmes *et al.*, 1977) and with high perceptual accessibility, whereas local colours require central vision to identify (Eisenberg and Zacks, 2016), and its perceptual accessibility is relatively low. Although peripheral vision and central vision have been demonstrated to play distinct roles in scene viewing (Larson and Loschky, 2009), it is still an open question regarding how different the local and global colour influence the reorientation efficiency. In addition, the indispensability of colours in improving the aesthetic quality of information visualizations motivates us to explore its role in attentional guidance and management.

2.6. Advantage of the eye-tracking method in examining attentional reorientation

Although behavioural recordings are the mainstream method that has been used in probing the attentional guidance effect of visual primes (Andersen and Maier, 2019; Biggs *et al.*, 2015; Soto and Humphreys, 2007), they do not allow a moment-by-moment examination of attention. By contrast, the measurement of spatiotemporal eye movement data shall be more diagnostic than traditional performance measures (e.g. response time and accuracy rate) (Kurzhals *et al.*, 2016). Although it is disputable when equating the focus of attention with the location of eye fixation, they are closely coupled in the visual search when the viewer is free to move their eyes (Findlay, 2004). Additionally, there is no appreciable lag between what is fixated and what is processed according to the eye-mind hypothesis[1] (Just and Carpenter, 1976). These back our presumption that eye tracking can provide a perspective from which we can observe how the attention is allocated over time. In fact, in the information visualization community, it becomes a conventional method using eye-tracking to uncover and explain the cognitive processes of the viewer through recording the distribution of visual attention (Kim *et al.*, 2012; Kurzhals *et al.*, 2013; Netzel *et al.*, 2014) and sequential characteristics of eye movement (Burch *et al.*, 2011; Jianu *et al.*, 2014; Peng *et al.*, 2021).

2.7. Hypotheses of the current study

Based on the related work review, we propose the following hypothesis:

H1: The target colour onto the preceding presentation will exert a priming effect on guiding the attentional reorientation upon arriving at the new presentation.

H2: Interferences will result from the invalid colour prime, which could be reflected by the lower visual search efficiency and sluggish attentional direction to the target object.

H3: More evident priming effect of global colours is expected to mitigate the attentional reorientation cost due to its facilitation on the pre-attentive processing of the search field.

H4: The attentional reorientation cost can be manifested by eye movement measures that correlate with the effort and efficiency for locating the target.

3. GENERAL METHODOLOGY

We conducted two experiments to examine the priming effect of colour on attentional reorientation in the series of *TemVis*. The first experiment is a confirmatory study of ***RQ2***, and the second experiment goes a step further to probe the ***RQ3***. Both exper-

1 It was revealed that visual attention moves slightly (around 250 ms) before the eye does (Deubel, 2008).

iments used the comparative research methodology between different colour priming conditions. The same paradigm that integrates the feature search task and keeping-track task was adopted in the two experiments.

3.1. Apparatus

Participants were seated in an 18 m^2 test room with two D_{65} fluorescent lights hung above and behind the participant. Direct and ambient sunlight was minimized by curtains, and the lighting condition was kept constant throughout the experiment and across participants. All experimental materials were presented by an HP 21-inch display with a brightness of 250 cd/m^2. Participants were seated approximately ranging from 600 mm to 750 mm in front of the terminal screen. We used a chin rest to keep their head position stable. During the test, participants' eye movements were recorded by Tobii X2-60 remote eye tracker. The reported binocular precision of the eye tracker is 0.34° (Tobii, 2014). Tobii studio version 3.2.3 was used to experiment programming, calibration, stimulus presentation and data analysis. Raw eye fixation data were filtered using the Identification by Velocity Threshold (I-VT Fixation Filter) with a 30°/s velocity threshold (Salvucci and Goldberg, 2000).

3.2. Participants

Two groups of participants were enrolled from the university bulletin system for two experiments (with a mean age of 26.52 years, SD = 2.74). Each has a Bachelor's degree or above in Engineering, Science or Psychology. Attributed to the positive correlation between education level and WM performance and the negligible WM performance difference between high education level groups (de Souza-Talarico *et al.*, 2007), the consistently high education level within participant groups rules out the confounding factor from inequivalent WM performance. They all have self-reported normal or corrected-to-normal vision and passed the Ishihara Colour Blindness Test before the experiment. The study compiles with departmental ethics committee regulations. Each participant was compensated 15 Yuan for each half hour of their participation.

3.3. Procedure

Before taking the experiment, each participant signed the informed consent form voluntarily. They were informed of the experimental procedure and the task. There were practice blocks (two blocks/10 trials for each experiment) before the test. No eye movement and behavioural recording were made during the practice. Following the practice is the five-point automatic eye tracking calibration by Tobii studio. Calibration was considered accurate when estimated fixation positions fell, on average, within around 0.4° for all points (as marked by Tobii Studio). Recalibration took place when accuracy fell below the established standard. In this condition, it is customary to assume that the spatial accuracy and precision of the quality of the eye-tracking data can be reported to be equal to the optimal specifications documented by the manufacturer: accuracy below 0.8° at the viewing distance of 600 mm $\sim$ 750 mm (Dalrymple *et al.*, 2018; Tobii, 2014). A 2-min break was set between units in both experiments to refuel the participant after a period of demanding attentional task (Rees *et al.*, 2017).

4. EXPERIMENT 1

The first experiment objective is to examine the priming effect of chromatic colours in mitigating the attentional reorientation cost in sequential *TemVis* that are simplified as alphanumeric arrays, and the interaction effect between *colour perceptual accessibility* (global vs. local colours) and *prime validity* (valid vs. invalid primes). Sixteen participants were recruited for the first experiment (five female, with a mean age of 24.7 years, SD = 2.8).

4.1. Method

Experimental design A 2∗2 within-subject design was employed with four colour priming conditions: the valid global priming, valid local priming, invalid global priming and invalid local priming condition. A neutral baseline greyscale condition was set for comparison. The prime validity was a within-unit factor, while the perceptual accessibility was a between-unit factor. Therefore, there were altogether five conditions, constituting three experiment units. Each *unit* contained 24 blocks, and each *block* further consists of five trials. Two successive presentations constitute one *trial*. In two colour priming units, valid and invalid priming trials were halved and randomized. The schematic framework of one experiment block is shown in Fig. 3.

Stimuli Design Each presentation has six patches arranged in three columns and two rows (323 px∗323 px for each). Each patch consists of 30 alphanumeric characters in a 6∗5 matrix. There were three sorts of alphanumeric characters in each patch: 26 *background distractors* and 4 *foreground distractors* (or 3 foreground distractors and 1 *target numeral*). There was only one numeral as the target in each display. The patch containing the target numeral was termed as the *target patch*. The dichotomy of global and local colour was according to Lloyd-Jones and Nakabayashi (2009). In the global colour priming unit, three sorts of characteristics within each patch were in the same hue but with different opacity. Whereas in the local colour priming unit, only foreground distractors and the target numeral were colourized, and the background distractors were in medium grey. The foreground distractors and target numeral were in the same greyscale, while the background distractors were in a lighter grey in the baseline unit. The sample display of each condition is shown in Fig. 4. Colours were picked from the CIE- L∗a∗b∗ space by maintaining a relatively equal

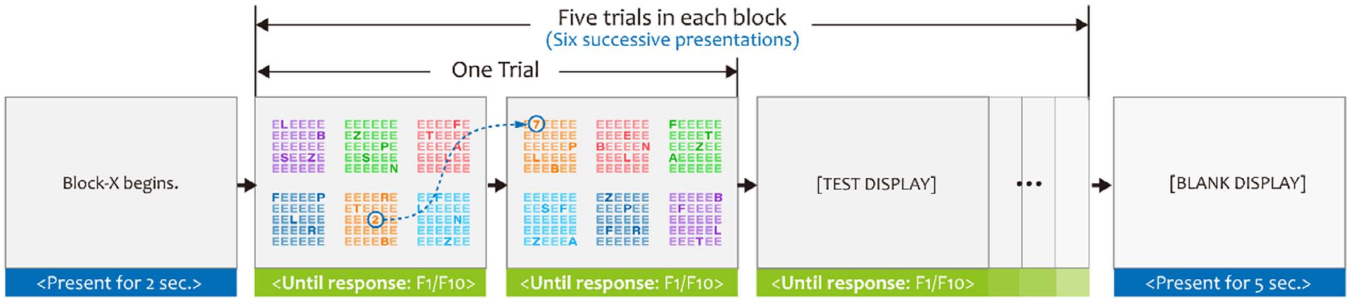


FIGURE 3. The schematic framework of one block in the global priming unit (the other two units have identical structures).

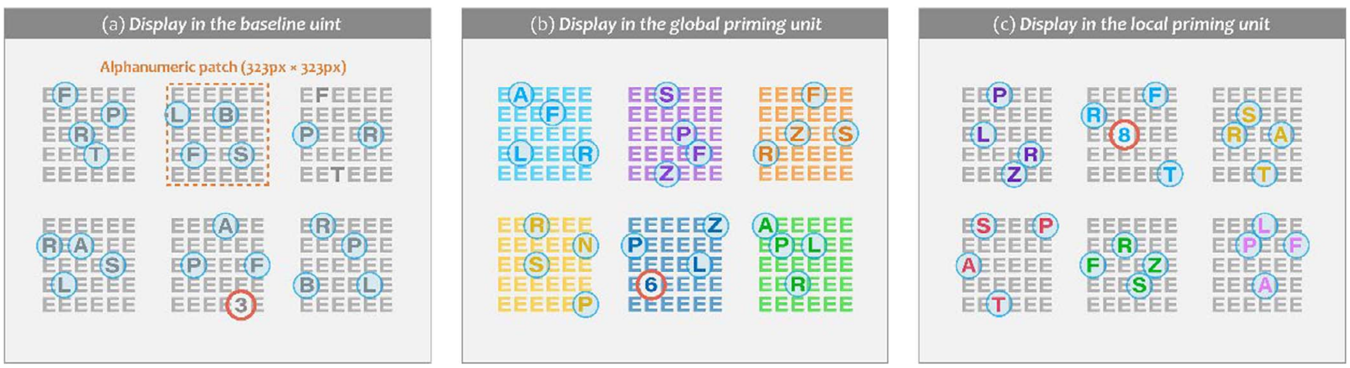


FIGURE 4. Sample display in the baseline, global priming and local priming unit. The foreground distractors were pointed out by blue bubbles, and the target numerals were in red circles. The letter 'E's were background distractors. The dotted box and bubbles are only for display.

level of colour difference between each pair (averaged delta ***E*** was >100 under the luminance condition during experiments). Controlling the colour perceptual difference was to eliminate the possibility of attentional guidance only by colour saliency (Mokrzycki and Tatol, 2011). The background colour of the display was light grey (Lab = 95.31/−0.49/0.29). We give the chromaticity setting of each sort of characters in three experiment unit in Table 1.

Background distractors were uppercase letter 'E'; foreground distractors were uppercase letters randomly picked from the set of {A, B, D, F, G, L, N, P, R, S, T, Z} and the target numeral was from 0 to 9. The foreground distractor set was defined based on the configural resemblance to the target numerals. The configural resemblance reduced the possibility of singleton search and added the difficulty of feature search. The target patch was pseudorandomly arranged, and its location remained different within one trial. Furthermore, the target patch location changed unpredictably across the display and with equal chance among the six positions (i.e. top-left, top-middle, top-right, bottom-left, bottom-middle, bottom-right). Participants were informed that the spatial location of the target patch is an uninformative prime or cue (Simons, 2000). In this way, they had no strategic reason to shift attention voluntarily to the primed location, so any attention shift should be codetermined by the task instruction and properties of the colour prime. In valid trials, the property of the patch colour was the same between two successive presentations, whereas colour properties differed in invalid trials. Valid and invalid trials were distributed randomly in the global and local priming unit. In Fig. 5, we give illustrations of one sample trial in four priming conditions.

Task setting and procedure Participants were instructed to finish a compound task of *feature search* for the only numeral among uppercase letter arrays and *track the numeral variation* across sequential presentations. It is pointed out that participants did not know the exact target digit in advance. Upon the onset of the new presentation, participants needed to search and identify the numeral through feature search among distractors. The target digit needs to be encoded into the WM for later processing. To make the variation judgment, a mental numeric magnitude comparison between the current and its preceding digit was engaged. The target numeral that has been encoded in the WM was retrieved for mental comparison. The variation can either be *going up* or *going down*, which correspond to *F1* and *F10* for the response. Participants were required to press the corresponding key immediately upon making the judgment. The presentation would get updated in response to the keypress, which manifested as the presentation replacement. Their

TABLE 1. Chromaticity and opacity setting of different characters in each experiment unit.

	Global priming unit	Local priming unit	Baseline unit
Target numeral	■	■	▲
Foreground distractor	■	■	▲
Background distractor	□	▲	△

The symbol ■ represents the chromatic colour in full opacity, while the symbol □ represents the colour in 70% opacity. The representational method applies to the baseline unit where only achromatic/greyscale alphanumeric characters were contained.

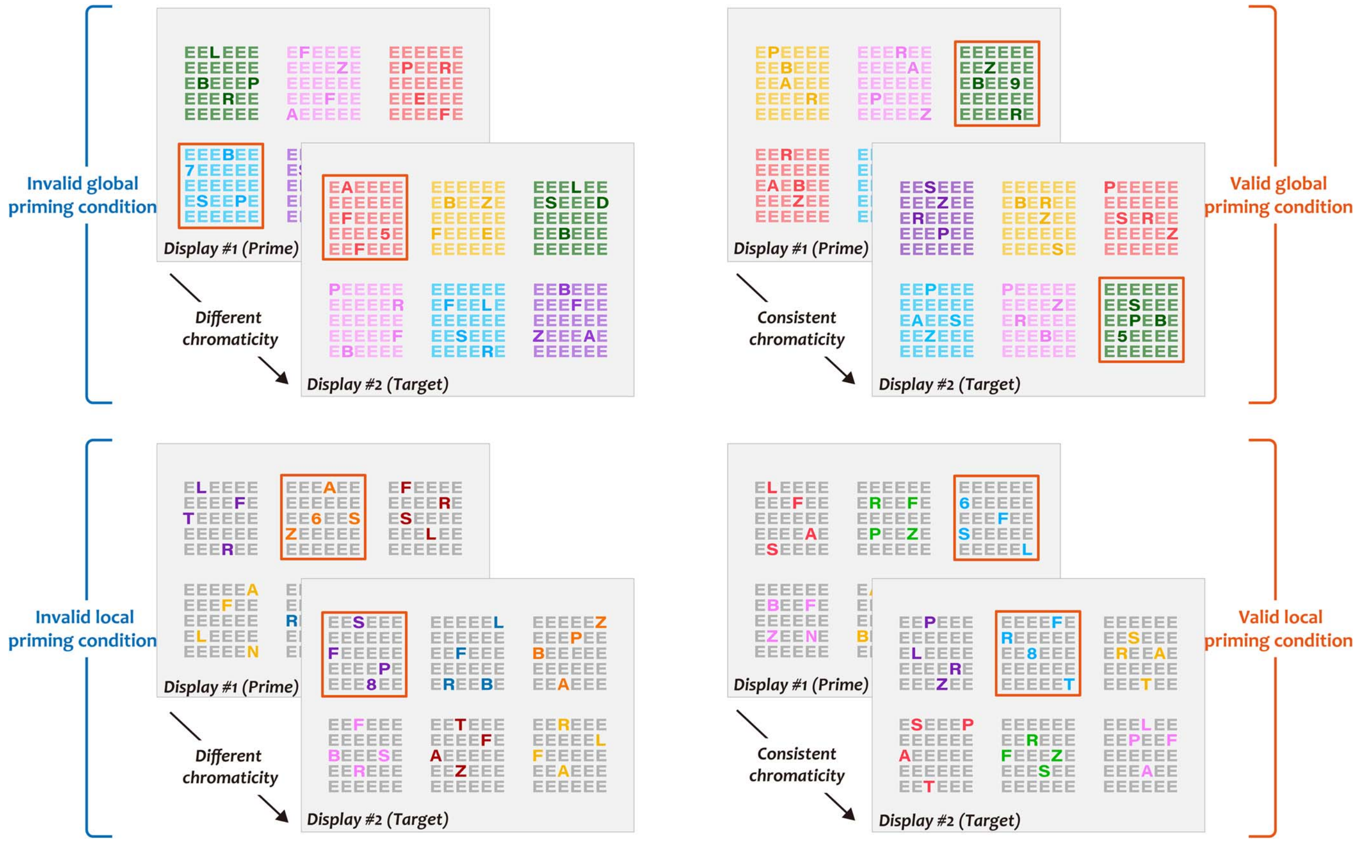


FIGURE 5. Sample trials in four chromatic priming conditions.

keypress latencies were recorded from the second presentation display in each block. A 5-s blank display was set between blocks to provide a short buffering time for the WM (Hillstrom, 2000). Orders of experiment units were counterbalanced between participants to guard against an ordering effect.

4.2. Dependent variables

Both behavioural and eye movement variables were recorded and included for analysis. Behavioural variables include *task completion time* (TCT) and *accuracy rate* (ACC) that were logged automatically by Tobii studio. Four eye movement variables were involved, including one direct measurement parameter—*time to the first fixation to the target patch* (TFF), and three indirect measurement parameters—*the ratio of TFF to total visiting time* (%TFF/TVD), *time to first fixation difference between the target and primed patch* (*delta* TFF) and *the percentage of invalid prioritization* ($p_{Invalid\ Prioritization}$). We listed the definition of each eye movement metric in Table 2. Among these four metrics, TFF and %TFF/TVD are complementary measures of search efficiency (Jacob and Karn, 2003). Meanwhile, the *delta* TFF and $p_{Invalid\ Prioritization}$ are indicators of search accuracy that refers to the ability to saccade the target rather than the primed region of interest (ROI) (Findlay, 1997). In other words, they reveal the interference effect intensity from invalid primes. Although previous studies have used saccadic measures (e.g. first saccadic latency, saccadic endpoints) (Findlay, 1997; Findlay *et al.*, 2001; Kowler *et al.*, 1995; Valuch

TABLE 2. Definition of eye movement metrics used in Experiment 1.

Metrics	Applicable conditions	Definition
TFF	All	Latency from the onset of the presentation until the participant visited the target ROI (not the individual digit) for the first time (also known as first visit latency).[a]
%TFF/TVD	All	The proportion of first visit latency to total visit time within the display during the presentation.[b]
delta TFF	Invalid conditions	Subtraction TFF (to target ROI) from TFF (to the invalidly primed ROI).[c]
$p_{Invalid\ Prioritization}$	Invalid conditions	The proportion of invalid prioritization counts to all invalid trials in two chromatic priming units.

[a]*TFF* is measured by unit of the patch instead of the individual alphanumeric digit. Given the reported precision of the eye tracker (0.34°) and the size of each digit (~50px∗50px), there exist some deviations between the recorded and actual fixation. Additionally, there might be slight trial-to-trial fluctuations in the recording deviation, making the digit-based recording hard to interpret in the current experiment.

[b]In the metric of *%TFF/TVD*, we used TCT as an approximation of total visit duration (TVD). For one thing, participants were instructed to press the button as soon as possible when they get the answer, and the response lag led by the eye tracker was negligible. For another, we can hardly dissociate the fixational visit time from response decision and manual execution time in the TCT.

[c]Negative values of *delta* TFF indicates that an earlier visit to the invalidly primed ROI than the target.

and Ansorge, 2015) as the proxy for search efficiency and accuracy, we chose the visiting time with one major consideration that concerns the splicing of saccades in the scan path. To be specific, different from the consistent initial central fixation point (began from the central marker) in previous studies, the starting fixation point may vary among all six patches on display. Considering the average amplitude between ROIs (20.61°, the calculation was according to the Euclidean distance between patches, frequencies of patch transition and the average viewing distance of 675 mm), the average saccade velocity of 30°–100°/s as suggested by Holmqvist and Andersson (2017, pp. 208–210) and the sampling rate of 60 Hz by the eye tracker, the scan path before entering the target ROI might consist of several saccades, thus making it pointless to adopt saccadic measures in this experiment.

4.3. Analysis

All recordings had a sampling rate above 85% as revealed by Tobii Studio, indicating the stability and high level of recording performance. Two types of ROI were predefined (350 px∗350 px for each) in each test display—one for the target patch and one for the primed patch. It is noted that the two ROIs overlapped in valid priming trials. Eye movement events outside ROIs are not considered. Data were excluded from analysis beforehand in trials fitting one of following four criteria: (i) trials with an incorrect response (i.e. a false judgment of the numeral variation); (ii) trials with no recorded gaze data; (iii) trials with TCT <100 ms (may due to accidental keypress); (iv) TCT falling over 2.5 standard deviation away from the mean value. We summarized all the removed trial counts following each criterion in each experimental condition in Table 3. Statistical analysis was conducted by SPSS 20.0. The confidence interval (CI) was set as 95% for all statistical analysis.

4.4. Results

A *post-hoc* power analysis using G∗power (Erdfelder *et al.*, 2009; Faul *et al.*, 2007) indicates that the achieved power was 0.883, with α at 0.05, and a large effect size assumed (i.e. effect size $f = 0.4$; Cohen, 1992). Both statistical power and the adequacy of sample size are validated. Summary results of TCT, ACC, TFF and %TFF/TVD in each condition are given in Table 4.

Task completion time and accuracy rates In general, averaged TCTs were shorter in valid priming conditions. The multivariate analysis of variance (MANOVA) results show that the main effect of *accessibility* factor is not statistically significant on TCT ($F_{(1,70)} = 1.229$, $P = 0.271$). However, the main effect of *validity* is significant ($F_{(1,70)} = 6.110$, $P = 0.016$) and the interactive effect between the two factors (*accessibility* ∗ *validity*) is marginally significant ($F_{(1,70)} = 2.849$, $P = 0.096 < 0.1$). According to the simple effect analysis, a significant within-unit difference is only found between the global priming unit ($M_{VG} = 1448.407$ ms, SD = 156.883; $M_{IVG} = 1776.928$ ms, SD = 382.976; $P = 0.04$). To compare the averaged TCTs in two valid priming conditions and the baseline condition, we adopted a one-way analysis of variance (ANOVA) with the *prime type* (i.e. valid global, valid local and no prime) as the independent variable. Its main effect is statistically significant on TCTs ($F_{(2,45)} = 4.470$, $P = 0.017$). The *post-hoc* analysis shows that the TCT difference between the valid global priming and the baseline condition reaches the statistically significant level ($P = 0.009$). In contrast, TCT between the valid local priming and baseline condition can be seen as statistically equivalent ($P = 0.758$). When comparing the TCTs in invalid

TABLE 3. Count of trials filtered out by individual exclusion criterion in each condition.

	Valid—global	Invalid—global	Valid—local	Invalid—local	Baseline
Criterion (I)	17 (1.77%)	24 (2.5%)	20 (2.1%)	28 (2.92%)	21 (1.09%)
Criterion (II)	4 (0.42%)	2 (0.21%)	7 (0.73%)	0 (0.00%)	3 (0.16%)
Criterion (III)	8 (0.83%)	4 (0.42%)	5 (0.52%)	2 (0.21%)	9 (0.47%)
Criterion (IV)	12 (1.25%)	15 (1.56%)	21 (2.19%)	5 (0.52%)	20 (1.05%)
	41 (4.27%)	45 (4.69%)	53 (5.52%)	35 (3.65%)	53 (2.77%)

Values in brackets indicate the percentage of rejected trials to total trials. Each of the four chromatic priming conditions has 960 trials (60 trials∗16 participants), while the baseline condition has 1920 trials (120 trials∗16 participants). Trials with TCT falling out of the range of ±2.5SD from the mean were all excluded since these trials were characterized by few fixations with long durations and great fixation dispersion during *post-hoc* inspection according to the gaze plot. These characters might reflect the mind-wandering during the attentional-demanding task (Krasich *et al.*, 2018).

TABLE 4. Summary results of TCT, ACC, TFF, %TFF/TVD in each condition.

	Valid—global	Invalid—global	Valid—local	Invalid—local	Baseline
TCT/ms	1448.4 (160.1)▲	1776.9 (380.0)■	1657.0 (209.0)	1729.3 (279.0)	1683.7 (338.1)
ACC/%	98.28 (4.66)▲	97.50 (7.81)	97.36 (3.96)	97.08% (6.30)■	97.17 (5.97)
TFF/ms	776.8 (86.6)▲	911.7 (104.3)■	834.7 (118.9)	838.4 (151.9)	793.2 (108.1)
%TFF/TVD	42.87 (4.40)▲	53.60 (6.91)	53.84 (9.22)■	52.62 (8.56)	51.68 (8.81)

The symbol ▲ marks the best performance, while ■ marks the worst performance in each dependent variable among five experimental conditions.

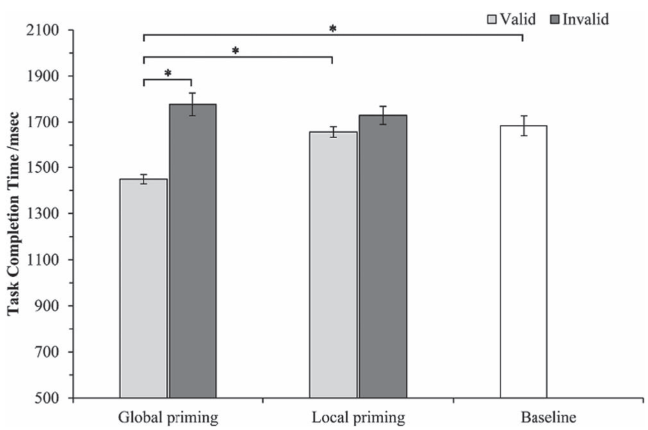


FIGURE 6. Task completion time in five experimental conditions. ∗ represents statistically significant at 0.05 level (P value $\leq$ 0.05).

conditions and the baseline condition, no significant difference is found ($P > 0.005$). We establish the TCT distribution and the paired comparison result between conditions in Fig. 6. In terms of the ACC, responses in the valid global condition had the highest ACC ($M_{VG} = 98.28\%$), whereas the lowest ACC is found when provided with invalid local primes ($M_{IVL} = 97.08\%$). However, no significant difference was found between units and within the unit ($P > 0.05$). We conjecture that there could be a ceiling effect since accuracies were above 95.00% in all experimental conditions (Salkind, 2012).

TFF and %TTF/TVD For clarity, we present the overall distribution of the two measures in Fig. 7. By and large, TFFs in valid priming conditions were shorter than those in invalid priming conditions. The valid global priming condition produced the shortest TFF and least %TTF/TVD, whereas the TFF was the longest when presented with invalid global primes. Besides, only a slight difference can be observed between valid and invalid local priming conditions. MANOVA results show that both the *accessibility* and *validity* factors have a significant effect on %TFF/TVD ($F_{(1,70)} = 6.224$, $P = 0.015$; $F_{(1,70)} = 5.656$, $P = 0.020$), and their interactive effect is also significant ($F_{(1,70)} = 8.925$, $P = 0.004$). Additionally, significant differences are found between %TFF/TVD *within* the global priming unit ($M_{VG} = 42.87\%$, SD = 4.40%; $M_{IVG} = 53.60\%$, SD = 6.91%, $P = 0.000$), and *between* the valid global and valid local priming condition ($M_{VG} = 42.87\%$, SD = 4.40%; $M_{VL} = 53.84\%$, SD = 9.19%, $P = 0.000$). Differences of %TFF/TVD between valid global and local priming condition ($F_{(2,45)} = 8.407$, $P = 0.000$), as well as between valid global and baseline condition ($P = 0.003$), are both significant based on the one-way ANOVA. However, no significant difference was found when taking two invalid priming conditions and the baseline condition for comparison ($P > 0.05$). Kruskal–Wallis H test reveals a significant effect of *validity* ($\chi^2 = 6.459$, $df = 2$, $P = 0.040$) on TFF, whereas no significant effect from *accessibility* ($\chi^2 = 1.053$, $df = 2$, $P = 0.591$) is found. Invalid global primes generated longer TFF than the baseline condition ($F_{(2,45)} = 3.300$, $P = 0.016$).

Delta TFF and $\boldsymbol{p}_{Invalid\ prioritization}$ To further examine the impact of colour perceptual accessibility on its priming effect,

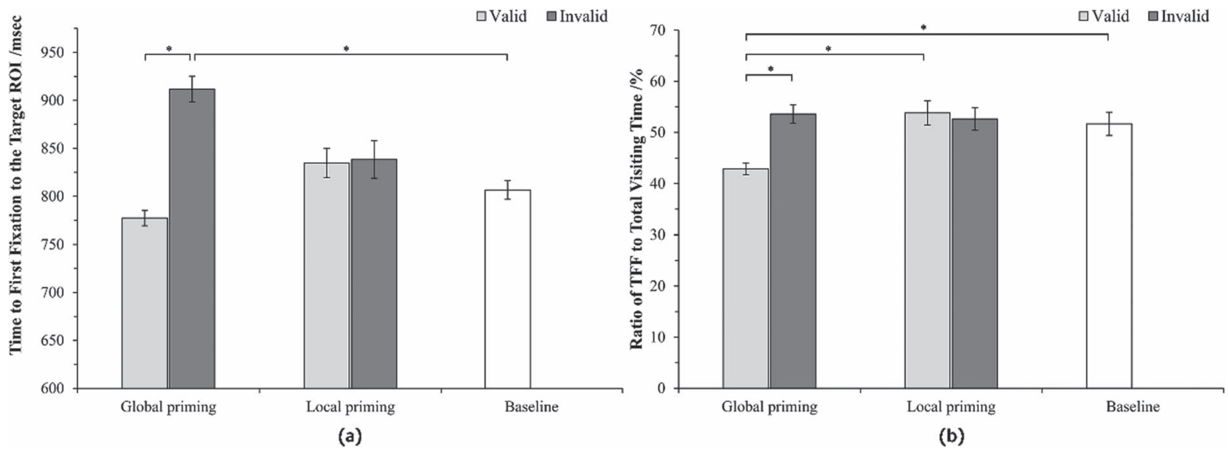


FIGURE 7. (a) TFF distribution across experimental conditions. (b) Distribution of %TFF/TVD across experimental conditions. Error bars indicate the standard error of the mean. ∗ represents statistically significant at 0.05 level (P value ≤ 0.05).

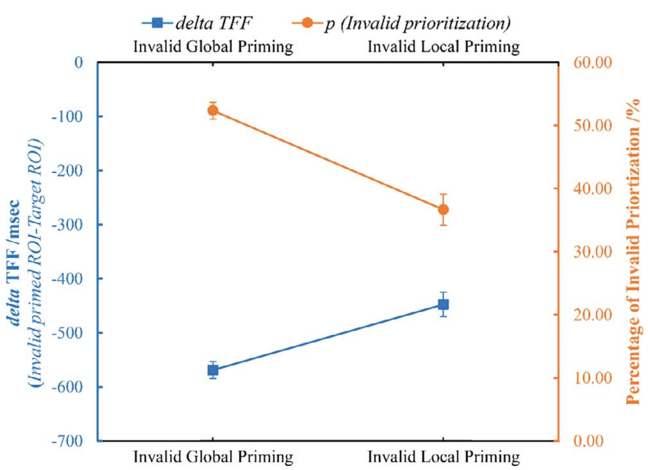


FIGURE 8. *delta* TFF and $\boldsymbol{p}_{\text{Invalid prioritization}}$ in two invalid chromatic priming conditions. Error bars indicate the standard error of the mean.

we currently focus on the invalid priming conditions. We discard trials where TFF to two designated ROI was <100 ms (Zhang *et al.*, 2018). As is shown in Fig. 8, the absolute value for delta TFF is larger under invalid global primes than the local primes, which is in accordance with the percentage of invalid prioritization. ANOVA reveals the significant main effects of accessibility on both *delta* TFF ($F_{(1,28)} = 4.828$, $P = 0.036$) and $\boldsymbol{P}_{Invalid\ prioritization}$ ($F_{(1,28)} = 18.529$, $P = 0.000$).

4.5. Discussion

In general, compared with the baseline condition, the overall TCT and the proportion of time that was spent on locating the target ROI to the overall completion time were shorter when valid colour primes were provided. It points to a faster targeting of the target region, which indicates that the viewer tended to show a better ability to select probable target regions when the prime was effective. Since the underlying factors of visual search slope (Treisman and Gelade, 1980) (e.g. the set size (Palmer, 1994), homogeneity of distractors (Bauer *et al.*, 1996a, 1996b), conspicuity between the target among distractors (Williams, 1966)) were controlled in the experiment, we infer that the saving of total completion time in the valid priming conditions should result from the reorientation process instead of the viewing process. The evidence supports our first hypothesis (**H1**). In addition, due to different configuration of chromatic patches between adjacent presentations, locating the target onto the new display requires participants to abandon the spatial associations that were established in the preceding display and to reset up their frame of reference when locating the object in the new search field (Mou *et al.*, 2008). Alongside the frame of reference reconstruction is updating the cognitive representation of the search field (Tolman, 1948; Zhang *et al.*, 2011). Therefore, we conjecture that the adding of colour as a prime can aid the construction of a new frame of reference and shrink the attentional reorientation cost following the presentation replacement.

Nevertheless, the facilitatory effect gets reversed when the prime and target was incongruent. The longer first visit latency and delayed completion time in invalid conditions lead us to surmise that the viewer may expend more time and effort locating the target. It reflects a low-efficient attentional reorientation following the replacement (**H2** is confirmed). Specifically, the impairment is remarkably evident in the invalid global condition (from the comparison between invalid global priming and neutral baseline condition). In contrast, no observable influence was found in the invalid local condition (from the comparison between invalid local priming and neutral baseline condition). Combining the *delta* TFF and $\boldsymbol{p}_{Invalid\ prioritization}$, we draw the inference that global colours bring more impacts than the local ones, and the priming effect from local colours is mild. Therefore, our third hypothesis (**H3**) is partially supported. To further examine and confirm the comparative effectiveness of global and local colours in guiding attention and mitigating the attentional reorientation cost, we performed the second experiment using real-world temporal data visualizations.

5. EXPERIMENT 2

The objective of Experiment 2 is to corroborate the finding in Experiment 1 that global colours exert a more intensive priming effect than local primes in guiding attention and mitigating the attentional reorientation cost in a real-world *TemVis*. We hypothesize that global prime should lessen the search difficulty and shrink the time and effort spent on directing attention to the target. A combined paradigm of visual search and keeping tracking task was adopted as the first experiment. Participants, procedures and recording methods are introduced in section 3. We used the same eye tracker as Experiment 1. Fifteen participants were recruited for the first experiment (seven female, with a mean age of 27.25 years, SD = 2.8).

5.1. Dataset

All data came from the WHO online database (http://apps.who.int/nha/data-base/QuickReports/Index/en). We chose Current Health Expenditure per Capita (CHE) data in ~60 countries from 2001 through 2016 for visualization. The data covers 60 countries from the European Union, African Union, Mideast, American and West Pacific region. Missing data from the dataset were extrapolated from nearby data. It was not important that the data be entirely accurate for our research purpose. Participants were required to complete tasks based on what they saw instead of on any preconceived notion.

5.2. Method

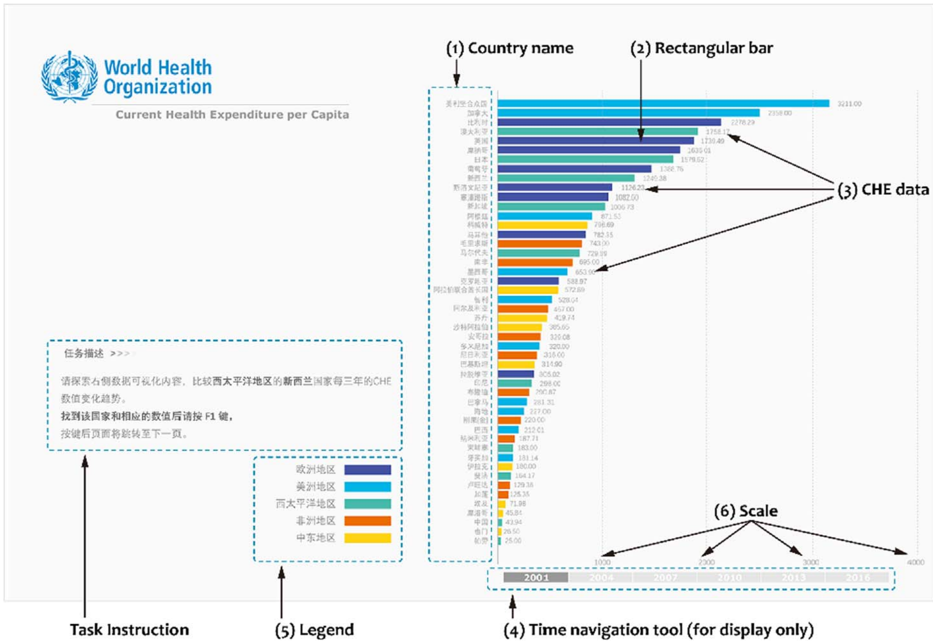
Experiment Design A within-subject design was employed with *colour perceptual accessibility* as the single factor. As is established in Experiment 1, two levels of accessibility were defined—the global colour and local colour. There were two blocks in each condition (unit), and each block consisted of five trials for measurement.

Stimuli Design We used the horizontal bar chart to represent the CHE data (every 3 years), and the CHE overtime was manifested by a series of presentations. As is shown in Fig. 9a, each bar chart consists of five components: (i) country name; (ii) bar; (iii) CHE data; (iv) time navigation tool; (v) legend; (vi) scale. Bars were colourized according to the geographical region where the country belongs. There were altogether five regions, and thus five distinct colours were displayed: $\mathbf{C_1}$ (R/G/B: 67/82/162, PURPLE); $\mathbf{C_2}$ (R/G/B: 0/180/230, BLUE); $\mathbf{C_3}$ (R/G/B: 72/186/172, GREEN); $\mathbf{C_4}$ (R/G/B: 237/108/0, ORANGE); $\mathbf{C_5}$ (R/G/B: 254/211/18, YELLOW). For example, Japan is in the Western Pacific region, so its colour in all visualizations is always the same ($\mathbf{C_3}$). The colours were determined considering the category effect on colour discrimination (Witzel and Gegenfurtner, 2014).

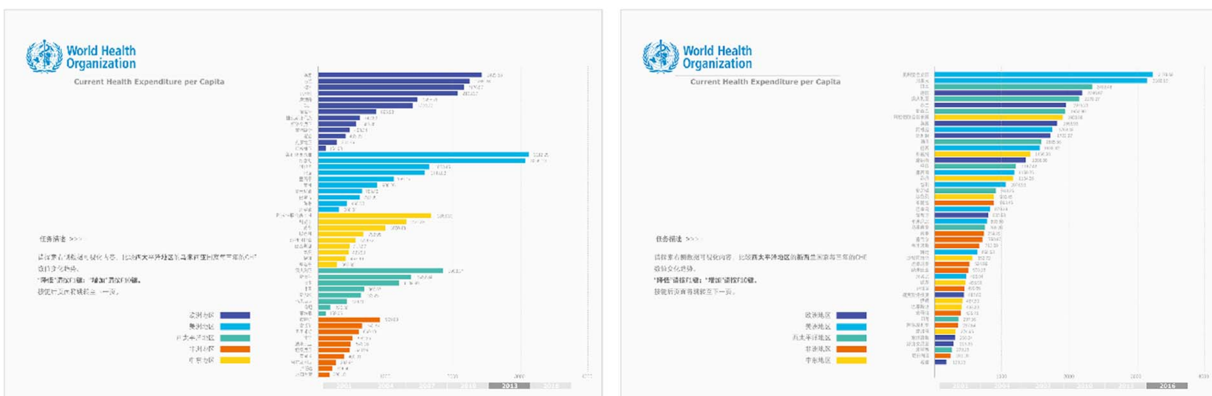
To echo the two perceptual accessibility conditions, we designed two formats of the horizontal bar chart. One is the region-grouped horizontal bar chart (RG-BC) and the other is the inverted-ordering horizontal bar chart (IO-BC). They correspond to the global colour and local colour, respectively. In RG-BCs, the order of regions (colour patches) is determined by the CHE summation of all countries within that region in that year. Within each patch, countries were ranked in the inverted order by values. Whereas all country data were placed in inverted order (with the country with the highest value on the top and lowest on the bottom) in IO-BCs (see Fig. 9b). Colours in RG-BCs were grouped and with high perceptual accessibility; however, the perceptual accessibility was weakened by scattered colourized bars in the IO-BC condition. We manipulated the raw data to ensure that the order of patches and the position of each country were not the same between successive visualizations. Visualizations were created by Tableau Desktop and embellished in Abode Illustration CC. Apart from different representation formats and task instructions, all other factors that might influence the task complexity (e.g. total country numbers and task type) remained the same throughout the experiment.

Procedure & Task setting Participants were initially briefed about the procedure and task type and then got familiarized with the experiment procedure through two training blocks (one for each condition). During the training, they were told about the design principle of two visualizations, and they got familiar with the geographical sections in both visualizations. Following the training was the test session, and the unit order was counterbalanced between subjects to avoid the ordering effect. All participants needed to finish six 7-point Likert scales, which aim at collecting subjective rating towards the effectiveness for perception, for navigation and aesthetics evaluation in each experiment unit after the test session. Cronbach's α of all three items was 0.926, 0.920 and 0.943, respectively. Bartlett's test of sphericity was significant (sig. = 0.000). The Kaiser–Meyer–Olkin measure of sampling adequacy was 0.776, suggesting a good structural validity of the scale.

The same task paradigm as Experiment 1 was used here. The only difference is that the target feature value (i.e. the target country name) was informed to participants in advance. One country was designated as the search and tracking target in each block (e.g. Morocco). Participants were required to track CHE data variations of the target country over triennial updates and judge whether there was an increase (pressing 'F1') or decrease (pressing 'F10') for each update. We give a sample task instruction as below. Each presentation terminated until the response.



(a)



Region-grouped horizontal bar chart (RG-BC)

Inverted-ordering horizontal bar chart (IO-BC)

(b)

FIGURE 9. (a) Components onto each test display. Task introduction and country labels were in simplified Chinese. (b) Sample display of region-grouped horizontal bar chart (on the left) and the inverted-ordering horizontal bar chart (on the right).

'Please explore the following temporal data visualizations by pages to track the variation of CHE value in the country of ***Morocco***. Make your judgment by pressing **F1** for an increase and **F10** for a decrease.'

5.3. Dependent variables

Behavioural and eye-tracking measures were recorded simultaneously during the test, while subjective ratings were collected after the test. Behavioural measures include TCT and ACC; eye-tracking measures include *total fixation counts* (TFC), *mean saccade amplitude* and *total scan-path length*. Given the sampling frequency (60 Hz) of the current eye tracker, we prefer to use *inter-fixation distance* (IntFix) and *the sum of inter-fixation distances* (Sum(IntFix)) as a coarse approximation of the saccade amplitude and total scan path length, respectively. The validity of approximation when using a low-speed and low-precision eye tracker has been confirmed by Megaw and Richardson (1979). We listed the definition of each eye movement metric in Table 5.

Among these eye-tracking measures, *TFC* can represent the fixation frequency and is a promising measure of search

TABLE 5. Definition of eye movement metrics and subjective ratings used in Experiment 2.

Metrics	Approximation	Definition
TFC	—	The sum of fixation numbers.
Saccade amplitude (SaP)	Inter-fixation distance (IntFix)	*SaP* is the distance travelled by a saccade from its onset to its offset; IntFix is the Euclidean distance between two successive fixations.
Scan path length (SPL)	Sum of inter-fixation distances (Sum(IntFix))	*SPL* is the sum of all saccade amplitudes in a scan path, while Sum(IntFix) is the sum of inter-fixation distances during scanning.
Perception effectiveness	—	To what extent the visualization support your finding the target information? (0 to 6 indicates *extremely useful* to *very bad*)
Navigation effectiveness	—	To what extent the visualization guide your attention immediately following the presentation onset and support you reinforce your search strategy? (0 to 6 indicates *extremely useful* to *very bad*)

efficiency (Goldberg and Kotval, 1999). *IntFix* reflects the size of the useful field of view and speed of information processing (Megaw and Richardson, 1979). Since the optimal scan path for a given task should be a straight line towards the target, longer scan path length may occur under ineffective guidance and poor-designed interfaces (Goldberg and Kotval, 1999). Compared with *SPL*, *TFC* and *SaP* can reflect the tempo-spatial distribution of fixations in greater detail, which further demonstrate the attention allocation along the time within the search field. Coordinate of fixations are determined against a 2D media coordinate system.

5.4. Analysis

All recordings were included due to the high sampling rate (all above 85%). The same eye-tracking event filter algorithm and data discard criteria were employed as Experiment 1. The removal process caused a total loss of 5.94% and 3.82% fixations in the IO-BC and RG-BC condition, respectively. Statistical analysis was conducted by SPSS 20.0. The CI was set as 95% for all statistical analysis.

5.5. Results

Post-hoc analysis using G∗power shows that the achieved power is 0.895, with α at 0.05 and effect size at 0.4, confirming the power of our statistical test and adequacy of the total sample size in Experiment 2. Two independent factors were extracted for analysis: the between-unit factor of *perceptual accessibility* and the within-factor of *presentation position* in the *TemVis* series (e.g. the first and second presentation in the series).

Task completion time and accuracy rates As the blue dotted lines in Fig. 10a, TCTs in the IO-BC condition were generally longer than TCTs in the RG-BC condition on all displays. Two-way repeated measure ANOVA (RM-ANOVA) reveals that the main effect of *perceptual accessibility* is significant ($F_{(1, 28)} = 25.922$, $P = 0.000$). Meanwhile, the *presentation position* also carries an significant effect on TCT ($F_{(2.876, 80.519)} = 33.038$, $P = 0.000$), as well as its interaction effect with perceptual accessibility ($F_{(2.876, 80.519)} = 35.452$, $P = 0.023$). Simple effect analysis indicates that in both conditions, the TCT difference between the first presentation and the five TCTs generated in the following five presentations were significantly different ($P < 0.05$). From the first to fifth presentation, the difference between the two conditions reaches the significant level or has a marginal trend towards significance. Moreover, no significant difference is found in the ACC between conditions, and they remained above 95.00% in all trials for all participants.

Sum of Inter-fixation distances A similar trend is found between the Sum(IntFix) (solid line) and TCT (dotted line) in Fig. 10a. Pearson correlation analysis confirms a significant correlation between these two variables ($r = 0.784$, P(two-tailed) $= 0.000$). Additionally, the total scan path length decreased from the first to the last presentation in both conditions; in particular, there was a remarkable drop between the first and second display in the IO-BC condition. Two-way RM-ANOVA reveals a significant main effect of *perceptual accessibility* ($F_{(1, 28)} = 15.548$, $P = 0.000$), presentation position ($F_{(2.103, 58.882)} = 17.788$, $P = 0.000$) and their interactive effect ($F_{(2.103, 58.882)} = 3.681$, $P = 0.029$). Simple effect analysis suggests that the Sum(IntFix) difference between two conditions achieve the significant or marginally significant level ($P < 0.1$) in all presentations in the *TemVis* series, except in the third display. Besides, the variation in RG-BC conditions is shallower, supported by marginal differences between the Sum(IntFix) in the first presentation and the following five ($0.5 < P \leq 0.1$). Meanwhile, a remarkable decrease was observed between the first and second presentation in the IO-BC condition ($P = 0.000$).

Total fixation count and inter-fixation distance As is shown by the orange lines in Fig. 10b, on average, fixation counts were smaller in RG-BCs than IO-BCs. RM-AVOVA confirms that the main effect of perceptual accessibility is significant

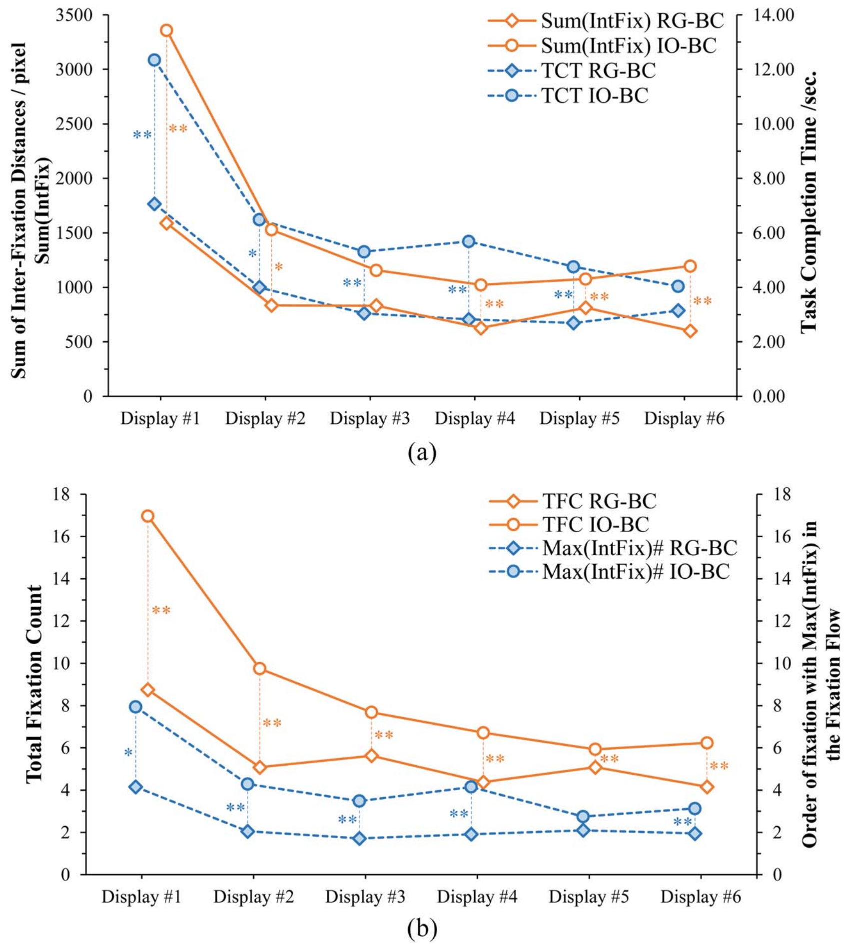


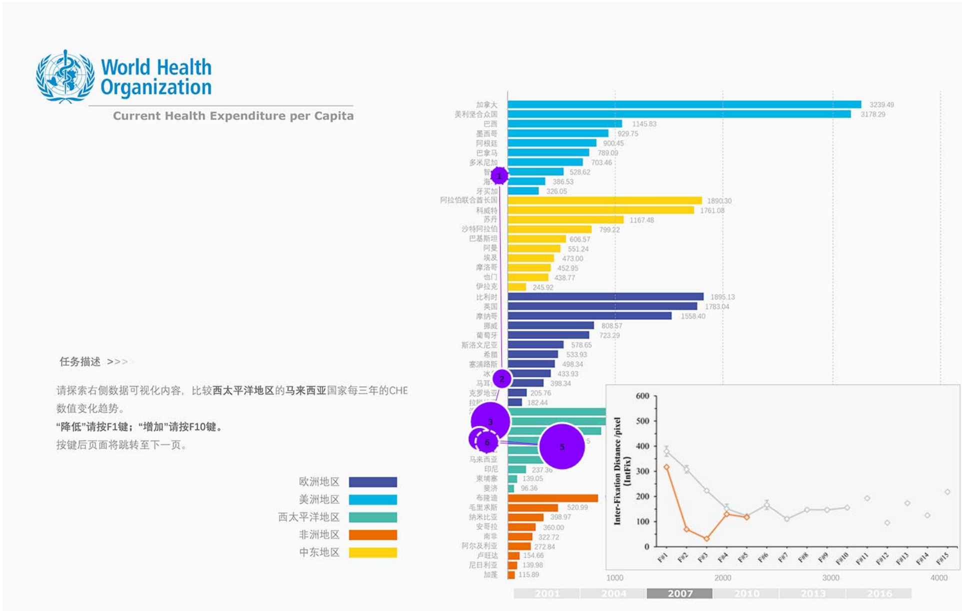
FIGURE 10. (a) Task completion time and the sum of inter-fixation distances in two units. (b) Total fixation count and the order of the fixation with the longest inter-fixation distance in the entire fixation flow. The fine lines establish the between-unit difference; the symbol of ** represents statistically significant at 0.05 level (P value ≤ 0.05), while * represents marginal significance (P value < 0.1).

($F_{(1, 28)} = 35.077$, $P = 0.000$), and the difference between units in any position also reaches the significant level ($P < 0.05$). A noticeable drop in TFC between the first and the subsequent five presentations was found in both conditions ($P < 0.05$). Then we extracted the fixation order with the largest inter-fixation distance in the fixation flow in two conditions (as the blue dotted lines in Fig. 10b). We find that the longest inter-fixation distance took place earlier in RG-BCs. RM-AVOVA results point out significant main effects of *perceptual accessibility* ($F_{(1, 28)} = 15.294$, $P = 0.001$) and *presentation position* ($F_{(1.746, 48.883)} = 7.578$, $P = 0.002$). For illustration, we establish one sample fixation plot from one participant in both conditions in Fig. 11. Some long saccades were detected from the beginning of the viewing process in the RG-BC. It seems that the participant directed his/her attention more efficiently to the target region, whereas long saccades were observed in the middle of the viewing process in IO-BCs.

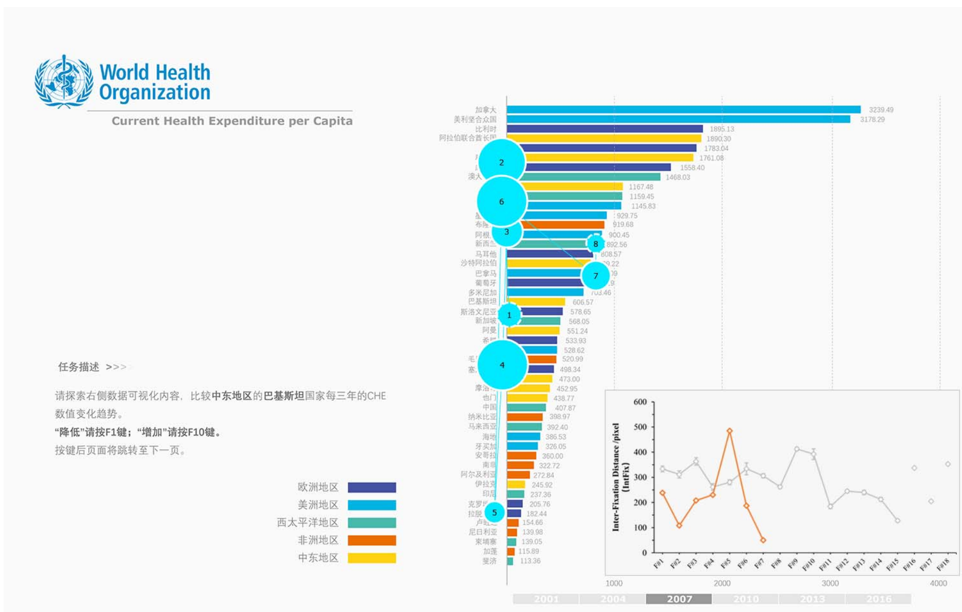
Subjective ratings Scores in all three ratings were higher with RG-BC than IO-BC (see Fig. 12). The higher rating concerning perception and navigation effectiveness coincide with the trend we have found in the aforementioned behavioural and eye-tracking measures. Additionally, RG-BCs were rated more aesthetically pleasing than IO-BCs. We conjecture that the viewer's perception and navigation performances affect their aesthetics evaluation towards the two visualization formats.

5.6. Discussion

Given the prolonged TCT, longer scan-path length and more fixation count recorded in the first presentation than the subsequent five presentations, we assume that the major contributor to the difference is a lack of prime in the first display of the *TemVis* series. Specifically, when landing on the first display, no prime in the WM could be utilized, whereas the priming colour feature that has been encoded into the WM can serve as attentional guidance in following presentations. Thus it is safe to conclude that the disparity confirms the existence and



(a)



(b)

FIGURE 11. Example of fixation plot (Participant #8) and inter-fixation distance distribution in the (a) RG-BC and (b) IO-BC condition. In the IntFix distribution plot (the line chart on the bottom right corner), grey lines show all participants' IntFix distribution in this presentation display, while orange lines show the IntFix distribution of Participant #8. The *x*-axis represents fixation#.; the *y*-axis represents its inter-fixation distance. Error bars indicate the standard error of the mean.

efficacy of priming effect on information search in the series of *TemVis*.

Owing to the relationship of two eye-tracking metrics with searching efficiency: total fixation count (negative relationship) and scan-path length (positive relationship) (Goldberg *et al.*, 2002; Mishra and Bhattacharyya, 2018), it is safe to conclude that searching efficiency was higher in the RG-BCs than IO-BCs. Furthermore, due to the high compatibility between the *fixation* detected by the I-VT algorithm and *cognitive fixation* when items are under processing (Galley *et al.*, 2015),

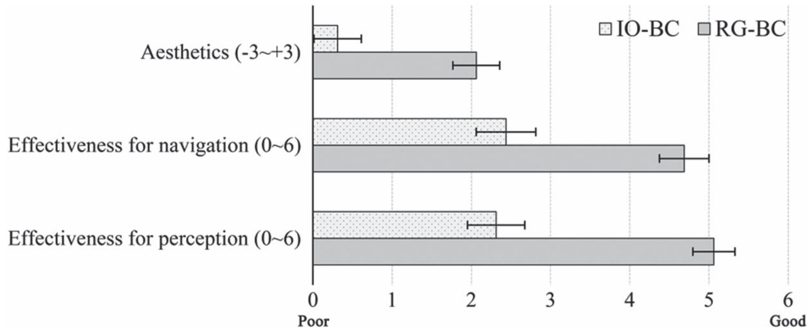


FIGURE 12. Subjective rating results for two visualization formats in Experiment 2. Error bars indicate the standard error of the mean.

it can be inferred that fewer items had been processed with the aid of global colour primes. Both the total fixation number and scan path length provide a granular perspective on the search efficiency. When breaking down the scan path, we gain a deep insight into how cognitive resources were distributed along the time.

In general, saccades with shorter amplitude represent local inspections of details, whereas larger saccades are assumed to reveal global overview scanning (Zangemeister *et al.*, 1995). Overview scanning and local inspection take place intertwiningly in the search field. We suppose that the overview scanning begins earlier in the global priming condition (RG-BCs) and takes up a more significant proportion in the whole viewing process. Conversely, more focal processing was involved in IO-BCs, since participants needed to inspect the country list until they found the target (Unema *et al.*, 2005). Considering there is always a trade-off between searching speed and accuracy, it is unreasonable to conclude which one is more favourable than the other, and it also exceeds the scope of the current study. However, the earlier initiation of overview scanning in the global priming condition, together with the evidence from the fixation count and scan path length, leads us to conclude that global colour primes are more conducive to improving the searching speed and efficiency of attentional reorientation.

Besides, some backtrack (e.g. Fixation#1→#2→#3 in Fig. 11b) and regression events (e.g. Fixation #3→#4 in Fig. 11b) are observed in IO-BCs. Some lookbacks can be detected if we carefully examine the fixation plot (refer to Holmqvist and Andersson, 2017, pp. 334–337, for distinctions among backtrack, regression and lookback). Fixations will move backwards to ROIs that were previously looked at in the local colour priming condition. Such a back-and-forth of scan path is responsible for a decrement in searching efficiency and also for the lower subjective ratings of perception and navigation effectiveness. According to the concept of inhibition of return, attention is less likely to be redirected to previously inspected areas within a transient temporal window (Posner *et al.*, 1985). Ideally, no refixation should be generated if perfect memory is engaged. Here, regressive fixations uncover the failure of WM (Gilchrist and Harvey, 2000). Therefore, it is human instinct to turn back their fixations during searching in the scene. However, more perceptible and salient colour primes (e.g. the global priming condition) can aid WM maintenance (Fine and Minnery, 2009). In other words, the enhanced visual–spatial memory encoding of colour sections will inhibit regressive fixations.

6. GENERAL DISCUSSION

Both behavioural and eye-tracking measures confirm the colour priming effect on guiding attention during reorientation when viewing serial visualizations. However, the priming effect demonstrated in the two experiments should belong to different priming categories.

Generally, the colour priming effect in the first experiment belongs to the *perceptual priming*, but both perceptual and *conceptual priming* are involved in the second experiment. With respect to the *perceptual priming* in Experiment 1, perception is facilitated by the congruence between perceived stimulus (i.e. the chromaticity of the fixated target) and the information stored in the perceptual representation system. Then, the stored content will non-consciously be retrieved and utilized to mediate the deployment of attentional resources during reorientation, which is manifested as the better searching performance in the priming condition than the neutral baseline. Our finding is also consistent with results in selective attention studies using event-related potentials, where attention to colour-cued stimuli will elicit an early anterior positivity (Kellenbach and Michie, 1996). Such positivity reflects the initial identification of stimuli (Hillyard and Münte, 1984) and the early selection of attentional focus (Broadbent and Gregory, 1964). In terms of *conceptual priming*, semantic and episodic memories of the scene are both involved. In Experiment 2, semantic-related regions, which are in the form of global colours with high perceptual accessibility, will be more attention-capturing and conducive to generating the high-efficient search strategy. Since visual features externalize semantic mappings (e.g. colourized bars are mapped to geographical regions in Experiment 2), we presume that the facilitation might partially result from the conceptual priming. In the meantime, perceptual priming should also work. Therefore, we contend that the colour priming effect found in Experiment 2 is the synergism of both perceptual and conceptual priming.

Due to the adoption of a compound task paradigm in both experiments, the viewer's attention can be influenced by the task instruction, which differs from the pure prime-driven guidance in previous experimental psychology experiments (Downing, 2000; Soto *et al.*, 2005). However, if goal-driven attentional shift always plays the dominant role, then little interference of invalid primes would be triggered in the experiment. However, it runs counter to results of lagged first visit latency towards the target in invalid priming conditions in Experiment 1. It leads us to presume that the task-driven guidance is interfered with by the invalid prime. If these two mechanism act in the same direction (i.e. in the valid priming condition), then a facilitation effect will be yielded. Regrettably, up to now, we cannot or even unlikely to figure out the time course of the interplay between task-driven and prime-driven attentional shifts.

Additionally, hierarchical encodings (i.e. the background distractor layer and foreground distractor layer) in Experiment 1 conform with human's hierarchical prioritization of visual analysis, that is, prioritizing the global level of analysis (Navon, 1977). Global prioritization can explain why colourized patch (in global conditions) can summon attention much quicker than the local colourized characters. The earlier initiation of large inter-fixation distance found in the global priming condition in Experiment 2 also supports the global prioritization effect. On the other side, global colours facilitate perceptual grouping (Roelfsema, 2006), which provides enhanced segmentation (of each patch) from neighbouring objects. By contrast, the segmentation is weakened by scattered colourized letters in local priming conditions. The segmentation boosts the pre-attentive processing of the entire search field by the onset of the new presentation, after which the viewer makes the attentional decision of where to allocate the first attention spot. Therefore, the segmentation enables the more speedy first attentional shift. The subsequent attentive processing of the local details is subservient on the pre-attentive processing result (Li and Dayan, 2006). In other words, the grouped units will then allow more efficient processing than did the corresponding processing of the fragmentary components in the local colour priming condition (Conci and von Mühlenen, 2009). Colour primes, especially global colours, can guide attention more effectively and maintain attention continuity after the view replacement. The attention continuity is a necessary but not sufficient condition for maintaining the *flow* when traversing between sequential visualizations (Csikszentmihalyi, 1988), thus leading to a higher perception effective rating and an alleviated level of getting-lost feelings in RG-BC visualizations.

Some limitations also exist in the present study. Firstly, a small proportion of oculomotor events might get lost during online recording due to the low sampling frequency of the current eye tracker. The I-VT filtering algorithm will also lead to a reduction in detected fixations and the total length of scan paths (Horley *et al.*, 2004). However, in this study, we focus on the *relative* eye movement features under different colour priming conditions. We believe the current recording equipment ensures the credibility of our study. Secondly, previous researchers reported a cumulative effect of primes (Maljkovic and Nakayama, 1994), but it is not observed in this experiment. It might be attributed to the combined effect of top-down and bottom-up priming effect in our task settings. We encourage future studies to examine the cumulative priming effects over time through the eye-tracking technique. Thirdly, there still an open question concerning the colour priming effect in sequential visualizations that updates at high rates. The current research findings inspire us to examine in a more dynamic environment in the next step.

7. CONCLUSION AND APPLICATION

Breaking the paradigm adopted by experimental psychology research, our experiment investigates the priming effect of colour in an integrated task paradigm, simulating the real-world scenario where the user needs to monitor the time-series information over time. In the first experiment, both behavioural and eye-tracking measures confirm an advantage for valid primes in guiding attention and reduce the attentional reorientation cost after view replacement. The benefit becomes more evident when the prime has higher perceptual accessibility. The finding is further supported by an empirical study using real-world temporal data visualizations. In both experiments, the searching efficiency was higher under the global colour prime. We speculate that the facilitation benefits from the pre-attentive processing of the entire search field if provided with global colours. Meanwhile, additional semantic information of the colour prime is inferred to activate the associative memory and facilitate the attentional reorientation process in Experiment 2.

The facilitatory effect of colours, particularly the global colours, in enhancing the search efficiency and shrinking the attentional reorientation cost after search field replacement can guide the design of spatio-temporal data visualizations. For example, in naval command-control combat interfaces, the location and direction of the fleet get updated in real-time, together with the specific parameters of each vessel. The situation map will refresh at a constant rate. To monitor and detect the potential risks, operators will continually inspect the map to grasp the overall situation. In this circumstance, providing global colours to the same fleet should contribute to faster target detection and detail inspection for each vessel. We can even speculate that the predominance of global colours will be more pronounced in scenarios featuring the high time pressure. Future research can focus on the efficiency of feature-primed attention in a more ecological environment.

Besides, the dynamics between the goal-guided and feature-primed attention can be a worthwhile research question with the combination of knowledge from neuroscience, experimental psychology and ergonomics.

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